

Topology-Aware Differentiable Triangle-Soup Reconstruction via Persistent Homology

Anonymous 3DV submission

Paper ID *****

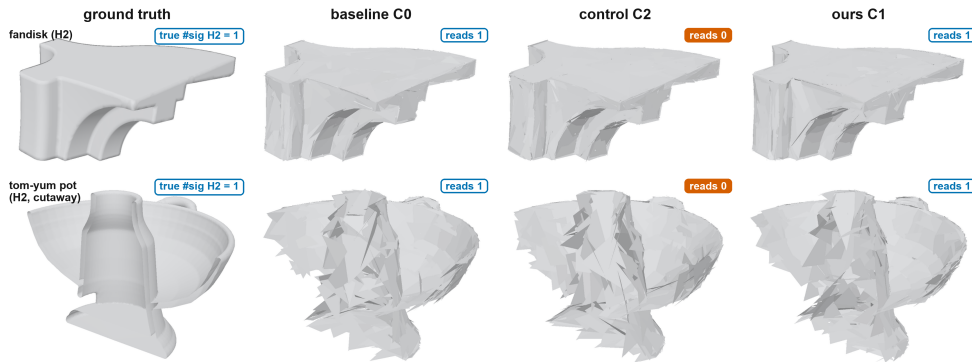


Figure 1. Geometry cannot carry the verdict — measurement can. Final reconstructions (seed 0), identical budgets: ground truth; baseline (C0); the norm-matched non-topological repulsion control (C2); the persistence loss (C1, ours) — equal-or-better Chamfer on every row (§5.2). A topological error need not be visible, so each cell is badged with its measured significant-feature count (blue outline = correct, filled orange = wrong; identical across seeds): the control erases both voids — and collapses a loop on bob (suppl. Fig. S8) — while the loss reads correct everywhere, cutting tail error up to $10.4\times$ (group means $1.5\times/4.9\times$; Tables 1–2).

Abstract

Differentiable triangle-soup reconstruction inherits a blind spot from its objective: photometric and geometric losses cannot see topology, so a reconstruction with a collapsed loop or a punctured enclosed void can score exactly as well as a correct one (on Chamfer-equal probes the diagrams differ $35\text{--}40\times$ in bottleneck distance). The standard implicit remedy — steer where the resampler spends its budget — does not repair this: in a controlled allocation study, a topology-informed prior is largely matched by an equally wide random one, and no prior shape repairs loops. We therefore move topology into the objective: a differentiable persistence term compares the evolving surface’s diagram, measured on live surface samples, to a fixed target; gradients flow through a pair-frozen backward re-expressing matched birth/death simplices as closed-form circumradii (exact for Gabriel simplices, verified per pairing), plus a recruitment term restoring the gradient optimal matching provably lacks when a target feature is missing; one ratio knob calibrates the loss against the photometric gradient, no curricu-

lum needed. Every claim passes a channel-controlled verdict: the loss must beat a norm-matched non-topological control through the identical gradient channel, at Chamfer parity. Under that rule the loss is topology-specific for enclosed voids ($4.0\text{--}7.9\times$ lower error) and — the class every allocation prior failed — for loops ($2.3\times$, zero phantom handles, while the control collapses one); loss and prior compose; component counts (H_0) are an honest null, matched by the generic control. The verdicts replicate without per-shape tuning on eight external genus-known meshes in two observable-class groups (pre-registered group means: loops $1.52\times$, voids $4.87\times$; one a raw artist mesh, ingested and certified; the one non-pass is a no-headroom null, its already-correct baseline left unharmed), degrading gracefully under sensor noise. Scope: all evidence is synthetic and single-machine — closed surfaces plus a first open-surface probe — with the target diagram known in advance, the regime where correction can be measured exactly; real scans and blind capture are future work. Prescription: correct topology in the loss, allocate wide, combine.

1. Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [47], periodically resampling under a fixed budget. Photometric losses are topologically blind: reconstructions differing in genus, connectivity, or enclosed voids can be indistinguishable to Chamfer and Hausdorff while persistence diagrams separate them 35–40× in bottleneck distance on Chamfer-equal probes (§2; suppl. Table S1). Fig. 1 previews the consequence: identical budgets, equal-or-better Chamfer, topology standing or falling with the objective.

A decade of connectivity-free reconstruction — point splats, Gaussians, and lately a return to triangle soups for their native graphics-pipeline fit [6, 21, 47] — has treated topology errors as symptoms of geometric under-provision: represent the surface better, or allocate the budget more wisely, and topology should follow. We tested the allocation half first (suppl. §A). It fails: a topological prior helps enclosed voids, but any equally-wide non-topological prior reproduces most of the gain, and no prior shape repairs loops — concentrating manufactures phantom handles, spreading never beats a random control.

The null result relocates the root cause: not where triangles are spent, but what the objective measures — a photometric loss is a local, pixel-space signal; a loop or an enclosed void is a global, categorical property of the surface. This paper therefore moves topology into the objective: a differentiable persistence term whose gradients act directly on features’ birth/death values rather than on where triangles respawn.

Three research hypotheses structure the study, each tested by a pre-designed condition:

1. RH1 (diagnosis): topology errors originate in the training objective, not in primitive allocation — tested by the allocation study above.
2. RH2 (specificity): a persistence term improves topology in ways no norm-matched non-topological regularizer through the identical channel can reproduce — the control C2 (§3.5).
3. RH3 (complementarity): objective and allocation are complementary channels, not substitutes — the composition C5.

Concretely, we contribute a matched persistence loss with a pair-frozen backward (§3): exact persistence forward (GUDHI alpha complex), backward through closed-form circumradii — Gabriel-exact for 100% of significant pairs in practice; a recruitment term for missing features: optimal matching yields zero gradient when a target feature has no live counterpart; recruitment attaches the nearest live bar, and removing it

collapses the loop-class win to baseline (§5); one calibrated knob: λ set by a gradient-norm ratio ρ , flat over a 10× range, no curriculum needed — an ablation, not an assumption; and channel-controlled evaluation: every claim must beat a norm-matched non-topological control through the identical channel (§3.5).

The loss proves topology-specific on the class no prior ever won — loops, zero phantom handles — confirming RH2; the channels compose (RH3); the H0 class is matched by the generic control, sharpening RH1: topological guidance earns its keep where geometry and topology decouple.

2. Related work

A decade of gradient-based reconstruction: four positions on connectivity. OpenDR [33], neural mesh rendering [27], SoftRas [32], DIB-R [8] and nvdiffrast [29] established gradient flow from pixels to geometry; reconstruction work since splits on what that geometry should be. (i) Implicit fields — learned occupancy and signed distance [35, 40], volume-rendered from NeRF [36] through NeuS [50] to Neuralangelo [31] — are watertight and topologically consistent by construction (the strongest counter-position), at field-evaluation and extraction cost. (ii) Fixed-connectivity templates deformed per scene [19, 49] keep manifoldness but freeze topology at the template’s genus. (iii) Differentiable iso-surfacing re-extracts the mesh from a volumetric scaffold at every step [37, 44, 45] (Shape-as-Points bridges point soups to watertight implicits [41]) — topological freedom at grid cost. (iv) Connectivity-free primitive soups — point splatting [51], patch atlases [17], Gaussians [28] and their surface-accurate variants [18, 24], and a 2025–26 return to native triangles (splatted [21], soft-connected [6], rasterized by DiffSoup [47]) — are the fastest and the easiest to budget, but even their surface-minded members regularize geometry (alignment, flatness, connectivity forces), never a measured topology. MeshSplatting, conversely, renders already-connected opaque meshes differentiably [20]: connectivity as input; ours a free soup with measured topology. (i) and (iii) solve topology representationally; (iv) had not addressed it. Cut by optimization philosophy rather than representation, the decade holds geometry-oriented objectives (every method above), sampling-oriented steering of where a fixed objective spends primitives (below), and an empty third family — topology-oriented objectives that measure the property and differentiate through it (suppl. Table S6). Prior work occupies every pairwise intersection of the section’s axes — explicit triangles, differentiable optimization, topology in the objective; this paper takes the three-way one, at the soup’s bud-

get and speed. We build on DiffSoup, whose in-loop prune/respawn resampler is the very mechanism our allocation study (suppl. §A) steered; here the resampler stays untouched and only the objective changes. Vertex-domain preconditioning [38] is complementary: it changes how vertices move, we change why.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper’s problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal by construction (0.916/0.628/0.977% per pair), the discriminating-dimension bottleneck separates correct from broken topology by $35\text{--}40\times$ — thin handle ≈ 0 vs .0302, bridged merge .0014 vs .0574, punctured void .0040 vs .1385 — and Hausdorff₉₅ even prefers the wrong candidate in two of three cases (suppl. Table S1). (2) For voids, a spread topological prior cuts the bottleneck 33–53% below baseline — but a width-matched random control recovers most of that gain, leaving a small, shape-dependent residual: the prior’s value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the non-topological wide control (.0267 vs the spread prior’s .0316), and the concentrated field manufactures phantom handles (4.4 significant loops vs the true 2); C5 below reuses the study’s strongest field (B4; suppl. §A).

Persistence in learning objectives. Persistent homology [13] with its stability guarantees [10] has been used as a trainable signal via differentiable sorting of filtration values [5], for topology-preserving image segmentation [9, 23], point-cloud and shape optimization and 3D volume prediction [4, 42, 48], and diagram-metric optimization with convergence analysis [7]; barcode-valued maps have since been given a general differential framework [30], and the per-step cost of persistence-guided descent attacked directly [39] — yet never as the objective of an explicit triangle-based reconstruction. Nearest our setting, concurrent work brings persistence to reconstruction at other, complementary entry points: an image-space barcode term for Gaussian splatting [46], homology-guided camera placement [15] or a genus-matched template [14] in mesh inverse rendering, and a connected-component constraint on implicits fitted to point clouds [25]. Partition our setting’s three requirements — (i) a topological quantity of the evolving surface measured differentially in the objective, (ii) image-based training through a renderer, (iii) a fixed primitive budget — and each neighbour satisfies at most one (suppl. §D.7): the image-space barcode term never measures the surface (i); the point-cloud pipeline forgoes renderer and budget (ii, iii);

the template line assumes the genus we measure (i). A head-to-head benchmark is therefore ill-posed (inputs, outputs, and cost regimes differ); our evaluation instead pits the loss against norm-matched controls through its own channel (§4). The nearest soup-side neighbour regularizes coherence with soft connectivity forces [6] — local continuity pressure, no measured topology; we replace that heuristic with the measurement. Our backward pass is in the persistence family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex’s birth/death as a closed-form circumradius — with three pipeline-specific ingredients. (i) Alpha-complex pairing over $\alpha \times \text{area}$ surface samples of a live triangle soup — not image grids or raw clouds — couples the diagram to the geometry actually being trained. (ii) A Gabriel-exactness gate: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — certifying the frozen gradient rather than assuming it. (iii) Recruitment for missing features: optimal partial matching has zero gradient exactly when a target feature is entirely absent, so we trade metric fidelity for a restored descent direction (§3), quantifying both its pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [2, 26, 43], and budgeted remeshing has a long lineage [1, 3, 11, 16, 22]; our contribution is orthogonal — the budget mechanics stay DiffSoup’s, only the training signal becomes topology-aware.

3. Method: a matched persistence loss for triangle soups

3.1. Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary + $0.8 L_1 + 0.2 \text{SSIM}$) [47], which we adopt unchanged: every ablation below isolates the topology channel, never photometric tuning. We add

$$L = L_{\text{photo}} + \lambda(t) L_{\text{topo}}(V), \quad (1)$$

where V are the soup’s vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed target bundle $\mathcal{T}$: the significant finite bars of the ground-truth shape’s diagram, computed once at the density M the loss will see (alpha values shift with spacing: density matching is a correctness requirement), diagonal-normalized, and locked for all live evaluations. Throughout, a bar is significant iff its lifetime exceeds the measurement floor $6 r_{\text{med}}(M)$ (r_{med}

248 = median nearest-neighbour sample spacing; derived
249 and validated in §6).

250 3.2. Live diagram and pair-frozen backward

251 At each refresh we draw M surface samples ($M=2048$
252 unless the measurement-floor rule of §4 dictates a
253 denser bundle) $X_j = \sum_i w_{ij} V[F(f_j), i]$ with faces
254 f_j drawn $\sigma(\alpha) \times$ area-weighted and barycentric weights
255 w_{ij} frozen; X is recomputed from live V every step, so
256 gradients reach vertices while the combinatorics stay
257 fixed. The alpha complex of X [12], computed exactly
258 with GUDHI [34], yields persistence pairs (birth sim-
259 plex, death simplex) per dimension (suppl. Fig. S9).
260 For a Gabriel simplex the alpha filtration value is its
261 squared circumradius, so each paired bar’s coordinates
262 are closed-form functions of a few sample positions, all
263 exactly differentiable:

$$264 \quad R_{\text{edge}} = \frac{1}{2} \|p_1 - p_0\|, \quad R_{\text{tri}} = \frac{\|u\| \|v\| \|u-v\|}{2 \|u \times v\|}, \quad (2)$$

$$R_{\text{tet}} = \|y\| \quad \text{with} \quad Ay = \frac{1}{2} \text{diag}(AA^\top),$$

265 where $p_0, \dots, p_k$ are the k -simplex’s sample positions,
266 $u = p_1 - p_0$, $v = p_2 - p_0$, and A stacks the rows $p_i - p_0$
267 (the circumcenter is $p_0 + y$). Edges carry H0 deaths and
268 H1 births, triangles H1 deaths and H2 births, tetrahe-
269 dra H2 deaths. We verify, per pairing, that the re-
270 expressed value matches the filtration value (relative
271 10^{-6}); a pair failing this Gabriel test is detached —
272 it keeps its (correct) value in the loss but propagates
273 no gradient, so an exactness failure can never inject
274 a wrong one. In practice the gate never fires: 100%
275 of significant pairs are Gabriel-exact on all six shapes’
276 clean clouds, and the per-refresh failure count is zero
277 at every one of the 5,500+ logged refreshes (all 2,400
278 noisy ones included, §5) — a constant pass rate, not
279 an aggregate. Stronger fallbacks (dropping pairs, sub-
280 gradients, locally raising M) remain available. The
281 pairing is treated as locally constant — the standard
282 pair-frozen device [5, 7] — refreshed every $K=10$ steps
283 and, necessarily, after every resampling event (which
284 replaces the primitive tensors wholesale); $K=10$ is an
285 amortization choice, not a tuned knob — it bounds re-
286 fresh cost (174 ms each; the fixed ≈ 46 s of §5) while the
287 pairing stays local.

288 3.3. Matching, diagonal pressure, and recruitment

289 Let D be the live significant bars and $\mathcal{T}$ the target
290 bars of one dimension. An optimal partial matching
291 (Hungarian on the standard augmented cost; equiva-
292 lently W_2) splits bars into matched pairs, unmatched

live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j) \text{ matched}} [(b_i - b_j^*)^2 + (d_i - d_j^*)^2] \quad (3)$$

$$+ \sum_{i \text{ unmatched}} \left(\frac{d_i - b_i}{2} \right)^2 + L_{\text{recruit}}.$$

293 The first term pulls matched features to their targets;
294 the second pushes spurious features to the diagonal; the
295 third exists because optimal matching has a provable
296 blind spot:

297 Lemma 1 If a target bar $t \in \mathcal{T}$ is unmatched under the
298 optimal partial matching, then L_{topo} without L_{recruit} is
299 locally independent of t : no component of its gradient
300 in V acts to create the missing feature.

301 Proof sketch. The unmatched target’s augmented cost
302 is its diagonal penalty $((d_t^* - b_t^*)/2)^2$, constant in the
303 live coordinates; every V -dependent term of Eq. 3 re-
304 ferences live bars only, and the matching is locally con-
305 stant in V (the pair-frozen regime) — no gradient term
306 is directed at t . $\square$

307 Recruitment restores a descent direction. Let $\mathcal{T}_{\text{miss}}$
308 be the unmatched target bars, processed in decreas-
309 ing persistence order, and let the candidate pool $\mathcal{A}$
310 hold every live bar not claimed by the matching — un-
311 matched significant bars and, crucially, sub-threshold
312 bars, which matching never sees. Each missing target
313 greedily claims its nearest pool bar, without replace-
314 ment, under the same squared birth–death metric as
315 the matched term:

$$L_{\text{recruit}} = \sum_{j \in \mathcal{T}_{\text{miss}}} [(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2], \quad (4)$$

$$k(j) = \arg \min_{k \in \mathcal{A}_j} [(b_k - b_j^*)^2 + (d_k - d_j^*)^2],$$

316 where $\mathcal{A}_j$ is the pool remaining after earlier recruit-
317 ments; recruited bars are excluded from the diagonal
318 term and back-propagate through the same circumra-
319 dius machinery. This deliberately departs from metric-
320 faithful W_2 ; §5 probes its location-blindness pathol-
321 ogy directly, and C6 shows the term is load-bearing.
322 Greedy rather than optimal assignment is deliberate:
323 in every logged regime the missing-target set is a sin-
324 gle bar (torus: one unreached target at every refresh,
325 every seed; §5), where the two coincide; when they di-
326 verge, decreasing-persistence order spends the pool on
327 the most significant features first.

328 3.4. Calibration instead of curriculum

329 $\lambda(t)$ ramps linearly from 20% to 50% of training, with
330 peak set once by a gradient-norm ratio at the first ac-
331 tive step: $\lambda_{\text{peak}} = \rho \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The sin-
332 gle knob ρ is dimensionless ($\rho=0.1$ throughout); the
333

tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, a constant- λ ablation matches the ramped version, and alternative ramp windows land within seed noise (suppl. Table S8) — calibration, not scheduling, makes the knob safe.

3.5. Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: $+$ a short-range repulsion on the same frozen samples, calibrated with the same ρ — the norm-matched control, the RH2 test (C2g: a gentler radius). C3: C1 without the ramp. (C4, curvature-weighted, is deferred; the numbering gap keeps condition tags comparable across studies.) C5: C1 + the best prior of our allocation study (the spread topological field B4; suppl. §A) — the RH3 test. C6: C1 without recruitment (pure matching + diagonal) — does the term matter in-loop, or only for cold-start repair?

4. Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (every-8th test holdout $\Rightarrow$ 21–63 training views): sphere and cube (void class H2, $\beta=(1,0,1)$, budget $N=1200$), torus (loop class H1, $(1,2,1)$, $N=700$), two spheres (component class H0, $(2,0,2)$, $N=700$), and genus-2 double torus as a stretch case $((1,4,1)$, $N=2000$). Budgets are the allocation study’s headroom points (suppl. §A) — tight enough that baseline topology is imperfect. Eight external, genus-known meshes probe generality beyond the analytic family, organized as the study’s two observable classes (§5.2): a loop group — bob (genus 1), rocker-arm (1), the CGAL eight (2) — and a void group — spot, fandisk, armadillo, horse (all genus 0), the tom-yum pot (3, below); sources, licenses and certificates: suppl. §G. Seven pass the mesh certificate below as-is; five further candidates failed it (open boundaries, non-manifold junctions) and were excluded, never repaired. Membership, observable class and density were fixed by the pre-registered staircase rule before any training (rocker-arm: torus precedent, H1-restricted at $M=2048$; eight: the pot’s, full $(1,4,1)$ bundle at $M=8192$). Scenes: the same pipeline (48 views, 200 px), unchanged class budgets (trio pilot: tails .043–.054, in the analytic band). The fourth, the tom-yum pot (Fig. 1, bottom row; aluminium showcase: suppl. Fig. S7), probes the step every real deployment needs: a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup (9 overlapping open shells, 232 non-manifold junction edges) with no trustworthy ground-truth topology — the same defect that excluded ShapeNet. An ingest-and-certify pass

(exact weld, offset-solidification into a thin closed shell, then certification by measurement: exact simplicial homology cross-checked against per-component Euler characteristics, both independent of the loss’s sampled alpha complex; watertightness; bit-identical rebuilds) yields one body of genus 3, $\beta=(1,6,1)$ exact, rendered and budgeted as a void-class shape ($N=1200$). This ingest pipeline is a test fixture here, not a claimed contribution. Where the other externals arrive clean enough to certify as-is, the pot’s topology exists only after ingest — a controlled real-world proxy, not a blind test: what real scans will need (mechanics and challenge inventory: suppl. §B, Table S4). The pot also exercises the floor rule of §6 prospectively: every H1 loop of the certified shell sits below the $6r_{\text{med}}$ floor at every working M , while the narrow flue mouth caps at small α — an H2 void on the pot axis, first clearing the floor at $M=8192$ (margin $1.20\text{--}1.23\times$, five sampling seeds). The pre-registered rule builds the bundle there (a floor-rule density deviation later repeated by eight; the loss samples at the same density): exactly one significant bar, stable to re-sampling ($\sim 10^{-5}$).

Protocol. All conditions share one implementation: C0 is the unmodified DiffSoup trainer; the loss enters as one additive term, and disabled it executes none of the added code (no renderer or resampler edits). 2,500 steps; resampling every 100 steps (untouched); $\rho=0.1$, ramp 0.2:0.5, refresh $K=10$, $M=2048$; decisive shapes (sphere, torus) and the double torus run 5 seeds (the former with the full condition set incl. C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2). (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss plan sees with i.i.d. Gaussian noise at every refresh: the plan decides on the noisy observation, the pulls apply to the true live positions. The torus restricts the loss to H1 (its own void bar is sub-threshold at $M=2048$ — a loss must not act on features it cannot reliably measure); the double torus likewise targets its two measurable loops of four. The generality wave runs C0/C1/C2, three seeds, each shape at its staircase-fixed bundle (bob: full loops+void; rocker-arm: H1 only; eight and the pot: $M=8192$).

Metrics and verdict. Primary: bottleneck to the target diagram in the discriminating dimension, tail-averaged (last three dumps, steps 2,300–2,500), seed-averaged (20k eval samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. Reporting policy: mean $\pm$ sd over seeds; Welch σ only where both arms ran five seeds; three-seed verdicts rest on disjoint per-seed ranges (suppl. Table S7). Verdict rule: C1 must beat C0 and

every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not one. Chamfer parity, formally: the passing arm’s seed-mean Chamfer must satisfy $\text{Ch}(C1) \leq 1.15 \text{Ch}(C0)$; the 15% headroom absorbs seed variance and nondeterminism (§7) and never did any work — every passing arm’s Chamfer was at or below baseline.

5. Results

A gate first: the loss alone (no photometric term, Adam on points) repairs all four controlled probes — a punctured void closes ($8.8\times$), a spurious handle dies at the diagonal, mis-spaced components separate ($1229\times$), a bridged merge with no significant live H0 bar (Lemma 1’s case) is severed ($450\times$, $\beta_0 1\rightarrow 2$); the predicted location-blindness did not materialize (suppl. §C, Fig. S5). Raw-cloud sanity toys; multipliers not comparable to training.

5.1. The C-matrix

Table 1 reports tail bottleneck (mean $\pm$ sd over seeds) in each shape’s discriminating dimension; Chamfer in the text where it matters; the full training trajectories for the two decisive shapes are in suppl. Fig. S6 (line = seed mean, band = seed min-max).

Voids (H2): topology-specific, large. Sphere $4.0\times$ (16.2σ Welch vs C0, five seeds), cube $7.9\times$ (three seeds: ranges disjoint, C1 .0064–.0086 vs C0 .0551–.0636), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is destructive, not merely weaker: it pins the void’s death at its repulsion equilibrium ($2.2\times$ worse, seed-identical) and erases the cube’s void outright; no tested strength reproduces any gain — $C1 < C0 < C2g < C2$ (§7; spread: suppl. Fig. S3; mechanism: §6).

Loops (H1): the channel’s distinguishing result. No prior shape won this class (suppl. §A). The loss cuts torus H1 error $2.3\times$ with zero phantom handles in all five seeds (count trajectories flat at the true $\beta_1=2$, suppl. Fig. S4), while the control collapses one loop by step ~ 800 and never recovers it. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not (mechanism: §6).

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error $9.5\times$, but the generic control matches it (.0008 vs .0007); the increment appears only in Chamfer (0.54 vs 1.94). Three channels agree: β_0 is carried by shell count and separation.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 ($7.6\times$ vs C0, $\approx 2\times$ vs C1 alone, $\approx 4.4\times$ below the prior alone), torus .0133 — with the best sphere Chamfer (torus .502, trailing only C3’s .495). Allocation and value-tuning are complementary, not substitutes.

Stretch case (genus 2): saturated, honestly. Every double-torus arm returns exactly .0264: the two tube loops are unexpressable at feasible budgets — the representation’s ceiling, not the channel’s (suppl. §B; floors: §6).

Ablations. The ρ -response is flat over a $10\times$ range (suppl. Table S8); C3 (no curriculum) $\approx$ C1 on both decisive shapes and off the analytic family (bob: C3 .0171 $\pm$.0026 vs C1 .0208 $\pm$.0008). Overhead, honestly: the loss adds a fixed ≈ 46 s per 2,500-step run (refresh 174 ms at $M=2048$ every 10 steps) — more than the 33–41 s photometric baseline on these deliberately tiny scenes, but the cost scales $O(M \log M)$ per refresh, not with scene or image complexity, so its share of a run whose photometric cost is T is $46/(T+46)$ — an arithmetic projection, $\approx 7\%$ of a ten-minute run — independent of image count and resolution; no parameters are added.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal) collapses the torus win entirely: .0411 $\pm$.0038, statistically baseline (0.6σ from C0, 12.4σ worse than C1) at baseline Chamfer, though both loops still exist (#sig H1 = 2 every seed; suppl. Fig. S6). The logs locate the mechanism: one torus loop is significant at every refresh, the other chronically below threshold (all five seeds, all 200 refreshes) — recruitment is the second loop’s sole gradient path; on this class it is the mechanism, not a cold-start device. On the sphere it never fires (0 of 200, all seeds): C6 is loss-identical to C1, and the 1.5σ tail gap (.0166 vs .0156) measures pure run-to-run noise (§7).

Sensor-noise stress (C7/C7h). With the plan’s cloud perturbed at every refresh ($\sigma = 0.5\%/1\%$ of the diagonal $\approx 0.5/1.0$ spacings, §4), the loss degrades gradually and never harms: torus .0210 $\pm$.0018 / .0259 $\pm$.0022 ($2.0\times/1.6\times$ below baseline, vs C1’s $2.3\times$), sphere .0235 $\pm$.0009 / .0328 $\pm$.0013 ($2.7\times/1.9\times$, vs $4.0\times$), at equal-or-better Chamfer, correct counts in every run (suppl. Fig. S6), zero Gabriel failures in all 2,400 noisy refreshes (internals: suppl. §B).

Table 1. Tail bottleneck to target (mean $\pm$ sd; lower is better); $\times$ = vs C0; verdict: C1 < C0 and < every control at Chamfer parity. Seeds: 5 (sphere, torus, double torus — saturated identically), 3 (cube, two spheres). C6 = no recruitment.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior	C6 no-recruit
sphere (H2)	.0623 $\pm$.0063	.0156 $\pm$.0013 (4.0 $\times$)	.1372 (worse)	.0151 $\pm$.0004	.0082 $\pm$.0007 (7.6 $\times$)	.0166 $\pm$.0005
cube (H2)	.0582 $\pm$.0046	.0074 $\pm$.0011 (7.9 $\times$)	.1346 (β_2 1 $\rightarrow$ 0)	—	—	—
torus (H1)	.0424 $\pm$.0031	.0180 $\pm$.0016 (2.3 $\times$)	.0457 (β_1 2 $\rightarrow$ 1)	.0155 $\pm$.0015	.0133 $\pm$.0004 (3.2 $\times$)	.0411 $\pm$.0038
two spheres (H0)	.0065 $\pm$.0008	.0007 $\pm$.0002 (9.5 $\times$)	.0008 (ties C1)	—	—	—
double torus (H1)	.0264 $\pm$.0000	.0264 $\pm$.0000 (saturated)	.0264 $\pm$.0000	—	—	—

Table 2. Generality wave, two observable-class groups (mean $\pm$ sd, 3 seeds; $\times$ = vs C0; group mean $\pm$ sd over member shapes, pre-registered). Seven of eight pass Table 1’s verdict rule; rocker-arm is a no-headroom null (counts intact in every arm). *seed-exact unmatched-target pins (§6). Per-seed values: Table S7; provenance, certificates, stair-cases: suppl. §G.

shape	C0	C1 loss	C2 control
bob	.0426 $\pm$.0012	.0208 $\pm$.0008 (2.1 $\times$)	.0437 (β_1 2 $\rightarrow$ 1)
rocker-arm	.0083 $\pm$.0004	.0087 $\pm$.0015 (0.96 $\times$)	.0113 (counts intact)
eight (genus 2)	.0249 $\pm$.0000*	.0162 $\pm$.0017 (1.5 $\times$)	.0249 (2 of 4 loops)
loop group mean (H1, 3)		1.52 $\times$ $\pm$ 0.55	
spot	.0521 $\pm$.0081	.0237 $\pm$.0000 (2.2 $\times$)	.0652 (β_2 1 $\rightarrow$ 0)
fandisk	.0538 $\pm$.0022	.0052 $\pm$.0008 (10.4 $\times$)	.0571 (β_2 1 $\rightarrow$ 0)
tom-yum pot	.0221 $\pm$.0003	.0057 $\pm$.0002 (3.9 $\times$)	.0223 (β_2 1 $\rightarrow$ 0)
armadillo	.0511 $\pm$.0000*	.0124 $\pm$.0001 (4.1 $\times$)	.0511 (β_2 1 $\rightarrow$ 0)
horse	.0408 $\pm$.0000*	.0109 $\pm$.0022 (3.8 $\times$)	.0408 (β_2 1 $\rightarrow$ 0)
void group mean (H2, 5)		4.87 $\times$ $\pm$ 3.17	

5.2. Generality: external genus-known meshes

Everything above is on one analytic family, so we repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, density fixed by the floor rule — on the eight external genus-known meshes of §4’s two groups, with no per-shape tuning. Seven of eight pass the verdict rule at equal-or-better Chamfer (Table 2; original-wave trajectories in suppl. Fig. S2); the eighth, rocker-arm, is a no-headroom null, not a counterexample (obs. 5). The control replicates its failure modes: all five voids erased (β_2 1 $\rightarrow$ 0, every seed; the pot at 1.36 $\times$ worse Chamfer), a bob loop collapsed (β_1 2 $\rightarrow$ 1), eight’s missing pair left missing.

Observations. (1) The H1 claim survives the family change: bob halves loop error, zero phantom handles every seed (suppl. Fig. S8), replicating torus. (2) The genus-2 ceiling was representational, not topological: on the fat CGAL eight, baseline and control read only two of the four certified loops (value pinned at the unmatched bound, seed-exact); the loss restores the missing pair (#sig 4/3/4) and cuts value error 1.5 $\times$ at 0.89 $\times$ Chamfer — the double torus’s saturation (§5.1) was the representation’s ceiling, not genus-2’s. (3) Sharp-featured CAD benefits most: fandisk’s 10.4 $\times$ exceeds every analytic-family effect. (4) Baselines can pin like controls: armadillo and horse read the correct void count but pin its value at the same

bound (.0511/.0408, seed-exact); the loss unpins both (4.1 $\times$ /3.8 $\times$) at better Chamfer. (5) No headroom, no effect — and no harm: rocker-arm’s one measurable loop is already baseline-correct (.0083 vs the .04-.05 band): §4’s imperfection criterion is unmet and the loss stays inert (0.96 $\times$, parity 1.00, counts intact everywhere) — the method’s honest boundary. (6) Count-level limits persist on sub-budget organic detail: spot’s baseline makes a phantom void in two of three seeds (#sig = 2, 1, 2); the loss clears two, retains one (1, 2, 1) — value repair reliable, count repair not guaranteed; its C1 tail pins at .0237 (sd 0), a milder floor of the double-torus kind. And the pipeline reaches raw artist content: the tom-yum pot (Fig. 1; suppl. Fig. S7), a raw Blender export certified rather than constructed and bundled at $M=8192$ by the same pre-registered rule (§4), cuts the void’s value error 3.9 $\times$ at Chamfer parity (0.99 $\times$), the study’s tightest seed spread (C1 .0056-.0059 vs C0 .0218-.0223); the control destroys the void (β_2 1 $\rightarrow$ 0, all seeds), pinned at the unmatched bound (.0223) — count, not value, is the verdict. Two caveats transfer: its baseline already reads the correct count (a value-accuracy win, like bob), and ground truth is certified, not constructed. Group statistic (pre-registered): loops 1.52 $\times$ $\pm$ 0.55 (2/3 pass), voids 4.87 $\times$ $\pm$ 3.17 (5/5); the passing wave spans 1.5–10.4 $\times$ (suppl. §G).

6. Mechanism and discussion

A local signal cannot carry a global property — sparse tugs can. A pixel’s photometric gradient reaches a few vertices; topology is global and categorical — a wrong genus is not approached smoothly; the blindness probes quantify the gap (§2) — no geometric proxy stands in. The persistence term supplies the measurement with a usable gradient: each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices — precise but sparse pulls, iterated by re-pairing; the resampling prior is the opposite — broad, indirect, budget-limited. The results sort along this axis: loops respond to value pressure (H1 win), wide shells to both channels (composition best), components to neither (H0 null) — and nothing responds where nothing is broken (rocker-arm’s no-

headroom null, §5.2).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface: the void’s death pins at that equilibrium (seed-identical) and drags while near-diagonal phantom bars accumulate (suppl. Fig. S1) and thin structures collapse (cube β_2 , torus β_1); the loss has no length scale — it pulls the death coordinate itself. The same gradient budget through the same channel without topological information is actively harmful: C1’s wins cannot be extra-regularization artifacts.

Measurability at the loss’s density. A persistence loss can only enforce features that clear the significance floor at its density M : the torus void and two of the double torus’s four loops do not, at $M=2048$. We treat this as a principle, not a nuisance: the bundle is built and audited at the same M , sub-floor features are excluded, and the double torus is evaluated against its two measurable loops — all the soup can express at feasible budgets.

The floor is quantifiable: significance means lifetime $> 6r_{\text{med}}(M)$, and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090→.0028 over $M=2048\rightarrow 20000$; ratio 3.16 vs $\sqrt{9.77}=3.13$; suppl. Fig. S10). The double torus’s tube loops live at lifetime .045–.046 on the clean surface: at $M=2048$ that is 0.83–0.85× the floor — invisible by a 15% margin, clearing it only from $M\approx 3\times 10^3$ — while at the evaluation density ($M=20000$, floor .0171) they clear it 3.1×; the binding constraint is the representation (the soup never expresses the tubes): the measurement floor recedes as $M^{-1/2}$, the representation floor is set by N and no M lowers it — the double torus sits on the second. The torus quantifies the noise interaction: its second loop clears the clean floor by only 1.33× — a margin live-cloud noise consumes, leaving recruitment as its only gradient path (C6), the loss reaching below its own floor (suppl. §C). Suppl. Table S3 tabulates counts across M ; the pot and the group wave ran the rule prospectively (§4; suppl. §G), validated in-loop (Table 2); adaptive density is the natural extension.

Prescription. Correct topology in the loss; allocate wide; combine — no schedule needed once the loss is calibrated against the photometric gradient.

7. Limitations

Synthetic, single-machine. All quantitative claims are on synthetic renders with known topology: an analytic family plus eight external genus-known meshes

in two groups (§5.2; one a certified raw artist mesh, the intended scan path), drawn from public repositories, not a curated benchmark suite. All surfaces are closed except a first open-surface probe, pre-registered and passed: two bowls cut from the sphere (openings $0.14R/0.50R$, certified disk topology) still read one significant void (the designed rim’s H1 never clears the floor); the loss cuts its error $4.2\times/4.1\times$ at better-than-baseline Chamfer (ranges disjoint); the control erases the wide bowl’s void in 2/3 seeds (suppl. §F). C7’s Gaussian probe bounds noise response, not scan realism (outliers, missing regions, registration error). Real scans remain future work: their boundaries add noise-born H1 bars (unlike a designed rim); bar-filtering is the near-term step.

Density floor. The loss cannot see features below the significance floor at its density (torus void; two of four double-torus loops at $M=2048$) — audited, but a real ceiling. Count-level repair of sub-budget detail is likewise out of scope (spot, §5.2).

Fixed target diagram. The target bundle presumes known ground-truth topology (a prior scan or template — not blind capture); template-free designs (total-persistence minimization, a target-genus penalty, class-level diagram priors) are untested.

Control strength. The repulsion control is aggressive (it destroys topology at calibrated magnitude); a half-radius variant is still 1.6× worse (sphere .0972 vs .0623): verdicts do not hinge on control tuning.

Nondeterminism. CUDA training is not bit-reproducible (borderline prune/split decisions flip under float drift at resampling); tail CVs run 7–10% for baseline and loss arms alike, a loss-identical re-run pair (sphere C6 $\equiv$ C1) landed 6% apart — pure noise; statements follow §4’s reporting policy.

Recruitment guarantees. The zero-gradient pathology it repairs is provable (Lemma 1); its own self-correction premise is empirical (suppl. §C) — pathological geometries could defeat it; a failure-case study is future work.

8. Conclusion

We moved topology from the allocation channel into the objective: a matched persistence loss with a Gabriel-certified backward and a load-bearing recruitment term, every gain beating a norm-matched control at Chamfer parity. The hypotheses hold (RH1 objective root cause; RH2 topology-specific; RH3 channels compose); verdicts replicate across two external groups and degrade gracefully under noise. Prescription: topology in the loss, allocate wide, combine; next: real scans. Reproducibility: code, scenes, bundles, logs and seeds ship on acceptance (nondeterminism: §7).

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